

Pseudopotentials for orbital-free DFT: Capturing nonlocality and correcting functional approximants

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Developing reliable pseudopotentials for orbital-free density functional theory (OF-DFT), especially for transition metals, remains a significant challenge. In this study, we provide a theoretical framework for analyzing pseudization strategies for OF-DFT calculations. From the analysis arises a proposed pseudization method which involves constructing local pseudopotentials by targeting existing Kohn-Sham DFT pseudopotentials through an optimized effective potential procedure. We produce four distinct sets of local pseudopotentials and evaluate their accuracy and transferability on the set of transition metal elements. Our results indicate a substantial improvement over previously available pseudopotentials. Although current OF-DFT functionals still only reach a qualitative accuracy for transition metals, our newly developed pseudopotentials provide a rigorous framework for further methodological advancements.

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I. INTRODUCTION AND BACKGROUND

Among all approaches to model materials at the atomic level, density functional theory as formulated by Kohn and Sham [1] (KS-DFT) stands out for its impact. Its most widely used implementations have so far been cited over 200 000 times [2]. The reason for such a success is that KS-DFT, in conjunction with functional approximations, offers a good compromise between accuracy and computational cost. However, independently from the choice of the underlying approximations, it becomes too computationally expensive when large, realistically sized systems (e.g., quantum dots, proteins, and complex materials interfaces) are targeted. The reason for this is the unavoidable costs of building and diagonalizing the KS Hamiltonian matrix. In particular, diagonalizations carry a cost of $O(N^3)$ number of operations, where N is measure of system size. There are many approaches that tackle KS-DFT's steep computational scaling [3–6]. Among them, a promising alternative is orbital-free DFT (OF-DFT) [7,8].

OF-DFT is based on the same theoretical footing as KS-DFT [1]. In particular, it borrows KS-DFT's expression for the total electronic energy given as a functional of the electronic

density:

$$E[n] = T_s[n] + E_H[n] + E_{xc}[n] + \int v_{\text{ext}}(\mathbf{r})n(\mathbf{r})d\mathbf{r}, \quad (1)$$

where $T_s[n]$, $E_H[n]$, $E_{xc}[n]$, and $v_{\text{ext}}(\mathbf{r})$ are the noninteracting kinetic energy, the Hartree (H) and exchange-correlation (xc) energy, and the interaction with the external potential (typically the Coulomb electron-nuclear attraction potential), respectively. The noninteracting kinetic energy is given in terms of KS orbitals (single-particle wave functions, hence the “s” subscript) as follows:

$$T_s[n] \equiv T_s[\{\phi_i[n]\}] = -\frac{1}{2} \sum_i n_i \langle \phi_i | \nabla^2 | \phi_i \rangle, \quad (2)$$

where n_i are occupation numbers. The electron density is simply

$$n(\mathbf{r}) = \sum_i n_i |\phi_i(\mathbf{r})|^2. \quad (3)$$

The Hartree functional is defined as the electronic Coulomb self-repulsion $E_H[n] = \frac{1}{2} \int \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} d\mathbf{r}d\mathbf{r}'$ and the xc functional $E_{xc}[n]$ is the energy term that is added to $T_s[n]$, $E_H[n]$, and $\int v_{\text{ext}}(\mathbf{r})n(\mathbf{r})d\mathbf{r}$ to recover the total electronic energy.

Following from Eq. (3), we can use the KS orbitals as variational functions to minimize the total energy, leading to the canonical KS equations

$$-\frac{1}{2} \nabla^2 \phi_i(\mathbf{r}) + v_s(\mathbf{r})\phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}), \quad (4)$$

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where $v_s(\mathbf{r}) = v_H[n](\mathbf{r}) + v_{xc}[n](\mathbf{r}) + v_{\text{ext}}(\mathbf{r})$ defining the H and xc potentials as functional derivatives, e.g., $v_X[n](\mathbf{r}) = \frac{\delta E_X}{\delta n(\mathbf{r})}$.

In OF-DFT, the noninteracting kinetic energy T_s is evaluated approximately by a pure density functional, which we call henceforth the kinetic energy density functional or KEDF. Using KEDFs allows the energy minimization of the functional in Eq. (1) to be based solely on the electron density, completely bypassing the KS orbitals, and generally leading to algorithms of linear scaling complexity [7]. The low computational scaling resulting from the employment of approximate KEDFs comes at an accuracy cost. In fact OF-DFT development almost exclusively focuses on development of KEDFs which are typically either semilocal [9–11] or nonlocal [12–14]. Although in this work we do not focus on KEDF development, we acknowledge that the accuracy of any result presented is heavily biased by the quality of the KEDF employed.

One thorn on the side of OF-DFT has persisted, namely, the handling of the electron-ion interaction. While this seems a nonissue [the $v_{\text{ext}}(\mathbf{r})$ is conceptually trivial], when approaching condensed phases the pseudopotential (PP) approximation is usually invoked [15–21]. Natural basis functions used to expand the KS orbitals for materials are plane waves $\chi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} e^{i\mathbf{k}\cdot\mathbf{r}}$, where $\mathbf{k}$ denotes a wave vector in the reciprocal lattice, and Ω is the real-space volume of the primitive cell such that the plane waves are orthonormal over Ω . To maintain a reasonable number of plane waves, a cutoff is imposed $|\mathbf{k}| < k_{\text{max}}$. Thus, PPs achieve two objectives: (1) at the expense of introducing nonlocality in the pseudopotential, they remove the need to explicitly consider core electrons as these are, in most applications, chemically inert; and (2) they eliminate the singularity of the Coulomb electron-nuclear potential at the nuclear positions [22,23].

Unfortunately, accurate and transferable nonlocal PPs (NLPPs), commonly used in KS-DFT, are not applicable to OF-DFT because they require KS orbitals to couple to the nonlocal part of the PP. In OF-DFT, as we have seen, orbitals are not immediately available.

Although we refer to recent reviews for additional details [7,8], in the past decades several attempts have produced effective local PPs (LPPs, hereafter) that have opened OF-DFT's applicability to main group metal elements, groups III-V elements third row and below, and a selected number of transition metal elements (such as Ag) [17,20,24–26].

The guiding principles to build LPPs have been to employ them in a KS-DFT calculation (i.e., with exact T_s) imposing it to reproduce the electron density, lattice constant, and even the band structure of solids [20,24,25]. For those elements for which this is possible, it is said LPPs can be employed in OF-DFT calculations. It is certainly surprising that band structures can be reproduced in pseudopotential calculations having only a local part. However, in many cases it was shown to be possible [27]. It turns out that LPPs can be successfully constructed for the main-group elements mentioned above (provided that the semicore states are excluded from the pseudization [19]) and also for a large array of transition metals (where instead semicore electrons are included).

The most accurate LPPs available to date for transition metal elements are the recently developed highly accurate LPPs by the Huang group [19]. The valence density of transition metals consists of contributions from both localized d states and delocalized s - p states. Existing KEDFs, which are largely based on the response of a homogeneous or nearly homogeneous electron gas, therefore struggle to describe these elements accurately [28]. In addition, accurate pseudizations of transition metals should account for semicore states that lie close in energy to the valence d bands and influence bonding and screening. Since OF-DFT represents $T_s[n]$ as a pure functional of the electron density, it lacks explicit angular-momentum dependence [21] and therefore cannot distinguish d -state character from s or p contributions. These combined factors explain why transition metals are a particularly demanding class of atoms for OF-DFT and motivate the construction of pseudopotentials that mitigate these limitations.

An inconsistency has persisted. That is, the LPPs have so far been developed based on comparison between KS-DFT calculations with LPPs and KS-DFT calculations with NLPPs. This modus operandi does not account for the important differences between approximate KEDFs typically employed in OF-DFT simulations and the exact $T_s[n]$ employed in KS-DFT. In this work we aim at addressing this inconsistency.

In fact, early on it was recognized that LPPs should be constructed specifically for approximate OF-DFT functionals so that they could not only provide a much needed smooth electron-ion interaction potential, but they could also provide partially a correction to the approximate KEDF potential [16]. These LPPs were soon abandoned [25] because the resulting LPPs had features that would induce numerical instability, would not lead to proper Coulomb decay of the LPP, and, most importantly, were found to yield mixed results for condensed phases. So it seems we are at an impasse, on one hand employing KS-DFT to build LPPs is likely to generate LPPs that are of little use in practical OF-DFT simulations where approximate KEDFs are used. On the other hand, building LPPs based on approximate KEDFs is associated with unmanageable side effects.

We also mention a notable recent approach, which involves evaluating directly the nonlocal part of the NLPP with orbital-free one-body reduced density matrices [29]. While the idea is promising, the resulting pseudopotentials are only accurate for a limited set of atoms, pointing to the fact that orbital-free formulations of reduced density matrices are often not N -representable [30].

Considering the pros and cons of the various LPP design principles, in this work we aim to produce a library of LPPs for transition metal elements that can be used in conjunction with approximate KEDFs. The design principles adopted in this work arise from a careful formal analysis of atomic pseudization (carried out in the following section) that can be summarized with three points: (1) LPPs should target an *existing* set of NLPPs, (2) LPPs should differ from the NLPP's local part by a correction that is spherical and short ranged, and finally (3) LPPs should be constructed in conjunction with approximate KEDFs.

II. AN IN-DEPTH LOOK AT PSEUDOPOTENTIALS AND PSEUDODENSITIES

In the PP approximation, the core electrons of each atom are represented by an effective nonlocal potential and the KS equations are only invoked for the valence electrons. The external potential becomes

$$v_{\text{ext}}(\mathbf{r}) \rightarrow v_{\text{NLPP}}(\mathbf{r}, \mathbf{r}') = v_{\text{loc}}(\mathbf{r})\delta(\mathbf{r} - \mathbf{r}') + v_{\text{nl}}(\mathbf{r}, \mathbf{r}') \equiv v_{\text{loc}}(\mathbf{r}) + \hat{v}_{\text{nl}}, \quad (5)$$

where we have indicated with a hat the nonlocal part of the NLPP.

NLPPs give rise to an electronic pseudodensity $n(\mathbf{r})$ that only reflects the density of the valence electrons outside the core radii of the atoms and is smooth inside. Such radii (NLPP cutoff radii, or $R_{\text{cut}}^{\text{nl}}$) are parameters of the NLPPs and their value is determined by convenience. Small $R_{\text{cut}}^{\text{nl}}$ will yield more accurate NLPPs but at the expense of needing to represent the more oscillatory components of the valence electrons.

The physical meaning of the nonlocal part of the NLPP is to maintain orthogonality between the valence and core orbitals. In KS-DFT, $\hat{v}_{\text{nl}}$ is usually given in terms of a limited number of spherical harmonic projectors centered on the atoms.

A. Nonlocal pseudopotential representations

Once a particular NLPP is introduced, the solution of the KS equations yields a pseudodensity $n(\mathbf{r})$. That same pseudodensity can be represented by many other NLPPs (hereafter indicated by $\hat{v}_{\text{NLPP}}$). The usual Hohenberg-Kohn map connecting external *potentials* with *densities* applies. However, this time we connect external, possibly nonlocal *pseudopotentials* with a target *pseudodensity*

$$\hat{v}_{\text{NLPP}} \longrightarrow n(\mathbf{r}). \quad (6)$$

Hohenberg and Kohn theorems for nonlocal external potentials due to Gilbert [31] apply and guarantee the existence of the above map, even though it is generally not bijective. The nonbijectivity is due to the fact that for a given choice of $\hat{v}_{\text{nl}}$ one can always find an associated $v_{\text{loc}}(\mathbf{r})$ such that the KS equations yield the target density $n(\mathbf{r})$. Therefore, there are multiple (potentially an infinite number) of $\hat{v}_{\text{NLPP}}$ that have nonzero nonlocal part and that yield the same electron density $n(\mathbf{r})$. We call the set of such NLPPs a *NLPP representation*. However, typically there is only one NLPP having the property that beyond $R_{\text{cut}}^{\text{nl}}$ not only the pseudodensity matches the all-electron density, but also the pseudo KS wave functions match the all-electron KS wave function for the isolated atom [32].

The Hohenberg and Kohn theorems guarantee that if $n(\mathbf{r})$ is noninteracting v_s representable, then there is one and only one purely local PP [hereafter denoted by $v_{\text{LPP}}(\mathbf{r})$] that, employed as external potential in the KS equations, yields the pseudodensity $n(\mathbf{r})$. Namely, the bijective map

$$v_{\text{LPP}}(\mathbf{r}) \longleftrightarrow n(\mathbf{r}) \quad (7)$$

exists. We depict the concept of NLPP representation and the maps in Eqs. (6) and (7) in Fig. 1.

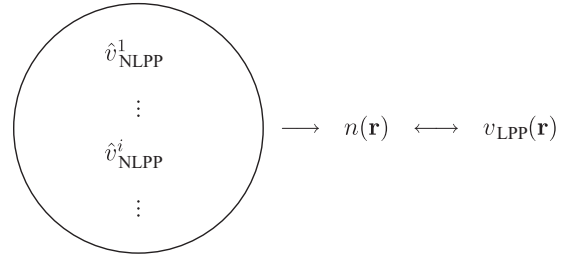


FIG. 1. Depiction of the maps in Eqs. (6) and (7) and the concept of NLPP representation (set of nonlocal PPs on the left-hand side) mapping to a target pseudodensity $n(\mathbf{r})$. The bijectivity of the map connecting the pseudodensity with the associated local KS potential $v_{\text{LPP}}(\mathbf{r})$ is indicated by a left-right arrow.

The relation between the local PP, $v_{\text{LPP}}(\mathbf{r})$, and any of the NLPPs, $\hat{v}_{\text{NLPP}}^i$, can be derived on the basis of the optimized effective potential method (OEP) [33,34], as we now explain.

B. Relating local and nonlocal pseudopotentials

We now seek to find the local PP, $v_{\text{LPP}}(\mathbf{r})$, that yields the target pseudodensity $n(\mathbf{r})$. We first define the relevant energy functionals, one for a NLPP and one for a LPP:

$$E_{\hat{v}_{\text{nl}}}[n] = T_s^{\hat{v}_{\text{nl}}}[n] + E_{\text{NL}}^{\hat{v}_{\text{nl}}}[n] + E_{\text{Hxc}}[n] + \int v_{\text{loc}}^{\hat{v}_{\text{nl}}}(\mathbf{r})n(\mathbf{r})d\mathbf{r}, \quad (8)$$

$$E[n] = T_s[n] + E_{\text{Hxc}}[n] + \int v_{\text{LPP}}(\mathbf{r})n(\mathbf{r})d\mathbf{r}, \quad (9)$$

where we have indicated with a superscript the quantities that are dependent on the choice of the purely nonlocal part of the NLPP, and we define the noninteracting kinetic energy in the NLPP case as

$$T_s^{\hat{v}_{\text{nl}}}[n] = -\frac{1}{2} \sum_i n_i^{\hat{v}_{\text{nl}}} \langle \phi_i^{\hat{v}_{\text{nl}}} | \nabla^2 | \phi_i^{\hat{v}_{\text{nl}}} \rangle \quad (10)$$

and the nonlocal energy $E_{\text{NL}}^{\hat{v}_{\text{nl}}}[n]$,

$$E_{\text{NL}}^{\hat{v}_{\text{nl}}} = \sum_i n_i^{\hat{v}_{\text{nl}}} \langle \phi_i^{\hat{v}_{\text{nl}}} | \hat{v}_{\text{nl}} | \phi_i^{\hat{v}_{\text{nl}}} \rangle = \text{Tr}[\hat{v}_{\text{nl}} \hat{\gamma}_s^{\hat{v}_{\text{nl}}}], \quad (11)$$

where $\gamma_s^{\hat{v}_{\text{nl}}}$ is the KS density matrix of the NLPP KS system.

Crucially, we must require that $E_{\hat{v}_{\text{nl}}}[n] = E[n]$, which delivers an important relation,

$$\begin{aligned} T_s^{\hat{v}_{\text{nl}}}[n] + E_{\text{NL}}^{\hat{v}_{\text{nl}}}[n] + \int v_{\text{loc}}^{\hat{v}_{\text{nl}}}(\mathbf{r})n(\mathbf{r})d\mathbf{r} \\ = T_s[n] + \int v_{\text{LPP}}(\mathbf{r})n(\mathbf{r})d\mathbf{r}, \end{aligned} \quad (12)$$

which can be rearranged to

$$T_s^{\hat{v}_{\text{nl}}}[n] + E_{\text{NL}}^{\hat{v}_{\text{nl}}}[n] = T_s[n] + \int \Delta v_{\text{LPP}}(\mathbf{r})n(\mathbf{r})d\mathbf{r}. \quad (13)$$

In the above, we defined $\Delta v_{\text{LPP}}(\mathbf{r}) = v_{\text{LPP}}(\mathbf{r}) - v_{\text{loc}}^{\hat{v}_{\text{nl}}}(\mathbf{r})$, as the potential that achieves a KS inversion [34–36] of the NLPP KS system onto a LPP KS system.

Taking a functional derivative yields a crucial result

$$\Delta v_{\text{LPP}}(\mathbf{r}) = \frac{\delta T_s^{\text{nl}}}{\delta n(\mathbf{r})} - \frac{\delta T_s}{\delta n(\mathbf{r})} + \frac{\delta E_{\text{NL}}^{\text{nl}}}{\delta n(\mathbf{r})}. \quad (14)$$

The above functional derivatives involve orbital-dependent functionals and thus can only be approached with OEP methods [33]. In the Supplemental Material [37], we show how to apply OEP in this setting and we also show that $\frac{\delta E_{\text{NL}}^{\text{nl}}}{\delta n(\mathbf{r})}$ decays faster than the Coulomb interaction, specifically like a dipole field outside $R_{\text{cut}}^{\text{nl}}$.

In addition to $\frac{\delta E_{\text{NL}}^{\text{nl}}}{\delta n(\mathbf{r})}$, let us also consider the $\frac{\delta T_s^{\text{nl}}}{\delta n(\mathbf{r})} - \frac{\delta T_s}{\delta n(\mathbf{r})}$, a term which is more challenging in nature. This is because the NLPP and LPP KS orbitals are generally different and would not yield the same kinetic energy unless the wave functions coincide beyond a radial cutoff [38]. We note that PPs are derived to maintain the Coulomb tail behavior beyond a certain cutoff radius, which we shall indicate by $R_{\text{cut}}^{\text{Coul}}$. Equation (14) shows us that the local parts of the NLPP should be equal to the LPP beyond the larger of the $R_{\text{cut}}^{\text{Coul}}$ among NLPP and LPP, ensuring that $\Delta v_{\text{LPP}}(\mathbf{r}) = 0$ beyond this cutoff radius.

Unfortunately, there is generally no guarantee that $R_{\text{cut}}^{\text{Coul}}$ for the LPP KS system will be small. However, early work [27], pioneering work by Mi *et al.* [20], as well as more recent studies from the Huang group [19], demonstrate that for several transition metals, $R_{\text{cut}}^{\text{Coul}}$ can be made as small as typical cutoff radii used in standard pseudopotentials.

Thus, we should expect to be able to construct LPPs for a significant set of transition metal elements. This is a quest we tackle head on in the next sections.

III. SIMULTANEOUSLY APPROXIMATING $\Delta v_{\text{LPP}}(\mathbf{r})$ AND THE OF-DFT OEP PROBLEM

We now change gears as we consider approximate KEDFs, indicated by $\tilde{T}_s$, typically employed in OF-DFT simulations. All expressions derived thus far apply only to systems where an exact T_s functional is used. Taking $\Delta v_{\text{LPP}}(\mathbf{r})$ from Eq. (14) from the previous section, we can now reformulate it to also correct for approximate KEDFs as follows:

$$\Delta v_{\text{LPP}}(\mathbf{r}) = \frac{\delta T_s^{\text{nl}}}{\delta n(\mathbf{r})} - \frac{\delta \tilde{T}_s}{\delta n(\mathbf{r})} + \frac{\delta E_{\text{NL}}^{\text{nl}}}{\delta n(\mathbf{r})}. \quad (15)$$

This definition clearly shows two goals for $\Delta v_{\text{LPP}}(\mathbf{r})$: (1) it includes the effect of the nonlocal part of the KS-DFT PP; (2) it corrects for the fact that KEDFs are approximate, making sure that the density (or the pseudodensity) of the OF-DFT system matches the one of the KS-DFT system. Thus, the result of approximating $\Delta v_{\text{LPP}}(\mathbf{r})$ is to simultaneously achieve the two goals.

To approximate it, we first impose it to be spherically symmetric around each ion, i.e., $\Delta v_{\text{LPP}}(\mathbf{r}) \equiv \Delta v_{\text{LPP}}(r)$, with $r = |\mathbf{r} - \mathbf{R}_I|$ for $\mathbf{r}$ in the vicinity of ion I , where $\mathbf{R}_I$ is the position of the I th ion. Then, we impose the following boundary conditions that ensure smoothness and short rangedness, with r_{cut} being a user-defined cutoff radius:

$$\begin{cases} \Delta v_{\text{LPP}}(r) = 0 & \text{if } r = r_{\text{cut}}, \\ \frac{d\Delta v_{\text{LPP}}(r)}{dr} = 0 & \text{if } r = r_{\text{cut}} \text{ and } r = 0. \end{cases} \quad (16)$$

We choose a polynomial ansatz

$$\Delta v_{\text{LPP}}(r) = a_0 + a_1 d(r) + a_2 d(r)^2 + a_3 d(r)^3 + a_4 d(r)^4 + a_5 d(r)^5, \quad (17)$$

where $d(r) = r - r_{\text{cut}}$. After applying the conditions above, we find a_0 and a_1 to vanish, and $a_2 = 3a_3 r_{\text{cut}} - 4a_4 r_{\text{cut}}^2 + 5a_5 r_{\text{cut}}^3$. The remaining three parameters (a_3 , a_4 , and a_5) are optimized minimizing the following cost function (exposing the dependence of the densities on the pseudopotentials):

$$\Delta n = \frac{1}{2} \int |n_{\text{OF}}[v_{\text{LPP}}](\mathbf{r}) - n_{\text{KS}}[\tilde{v}_{\text{NLPP}}](\mathbf{r})| d\mathbf{r}, \quad (18)$$

where

$$n_{\text{OF}}(\mathbf{r}) = \arg \min_n \left\{ \tilde{T}_s[n] + E_{\text{Hxc}}[n] + \int v_{\text{LPP}}(\mathbf{r}) n(\mathbf{r}) d\mathbf{r} \right\} \quad (19)$$

and $n_{\text{KS}}(\mathbf{r})$ is the KS-DFT pseudodensity arising from a particular choice of nonlocal PP. The minimization effectively, yet approximately, implements OEP because the loss function is minimum when the OF-DFT density is close to the KS-DFT pseudodensity. Important aspects to note are that $\Delta v_{\text{LPP}}(\mathbf{r})$ is (1) short ranged around each ion, as it is only nonzero for $0 \leq |\mathbf{r} - \mathbf{R}_I| \leq r_{\text{cut}}$; (2) of spherical symmetry around each ion. Thus, the resulting LPP, given by $v_{\text{loc}}^{\text{nl}}(r) + \Delta v_{\text{LPP}}(r)$, maintains the correct Coulomb decay in agreement with the target NLPP.

The next task is to choose an appropriate valence configuration for a pseudoatom. This is a complex task and has been addressed extensively in the pseudopotential literature. In this work, we simply adopt the valence configuration and the number of valence electrons from the corresponding target KS-DFT pseudopotential. As shown in Table S1 [37], for some elements the number of valence pseudoelectrons (zval) differs between the two types of pseudopotentials considered. This difference arises from variations in zval among the corresponding KS-DFT pseudopotentials. We also note that there is freedom in choosing the cutoff radii (r_{cut}). We use the value of the largest of the $R_{\text{cut}}^{\text{nl}}$ of the target NLPP (see Supplemental Material Table S1 [37]). We will see in the results section that this choice might not be the most optimal given the restrictive, polynomial ansatz for $\Delta v_{\text{LPP}}(r)$. We acknowledge that the choice of cutoff radius will deserve further consideration in future works.

Regarding the choice of polynomial ansatz for $\Delta v_{\text{LPP}}(r)$, after testing, we found that a fifth order provides a good balance between accuracy of the LPPs and computational cost of the LPP optimizer. Reducing the order of the polynomial would result in significantly lower quality LPPs, and increasing the order would result in significant computational cost and convergence complexity for the optimizer.

Finally, we note that the proposed pseudization results in a overall smooth PP. The short-ranged polynomial is necessarily smooth because the high-order terms (beyond fifth) are not included. The Coulomb tail is recovered from the LPP of the target NLPP. Thus, the new LPPs are designed to be numerically well behaved.

IV. COMPUTATIONAL DETAILS

The KS-DFT calculations were performed using the Python package *qepy* [39], which is interfaced with quantum espresso (QE) [40]. All simulations employed a Monkhorst-Pack grid [41] of $11 \times 11 \times 11$ k points and a plane-wave cutoff energy of 680 eV for the wave functions (four times as much for the electron pseudodensity). We chose two non-local, ultrasoft PPs, namely, GBRV [22] and PSL [42] due to their ability to retain accuracy across the periodic table while maintaining a workable plane-wave cutoff. The KS-DFT calculations were run with these PPs while the OF-DFT calculations were run with either only the local part of the PPs or with the additional $\Delta v_{\text{LPP}}(\mathbf{r})$ correction discussed before.

The OF-DFT calculations were carried out using the *dftpy* Python package [43] with a plane-wave cutoff energy of 1600 eV. To compare to existing LPPs for OF-DFT simulations, we employed the high-quality local pseudopotentials (HQLPP) of recent conception [19].

All calculations employed the local density approximation (LDA) xc functional proposed by Perdew and Zunger [44]. Therefore, target results in this work are given by pseudopotential KS-DFT simulations employing the LDA xc functional. We also tested our pseudization procedure using Perdew-Burke-Ernzerhof (PBE) [2] for Ag recovering a similarly accurate pseudization as LDA (see Results section). To test the effects that different xc functionals may have on the pseudization procedure, we included Tables S10–S12 [37] collecting equation of state (EOS) parameters for Ag using PBE. Testing other xc functionals will be matter of future work.

Unless otherwise stated, bulk calculations are carried out on the primitive cell of the most stable phase, the equation of state (EOS) curves were generated by varying the lattice parameter from $0.9V_0$ to $1.2V_0$, where V_0 is the equilibrium volume. From these curves, the minimum energy (E_0) and bulk modulus (B_0) were obtained by fitting the energy versus volume data to Murnaghan's equation of state [45]. We further assess the transferability of the new pseudopotentials using a nonlocal KEDF, revHC [46], which is based on the HC functional [12], for the EOS parameters.

Phonon spectra calculations were performed using experimental lattice constants and different methods for KS-DFT and OF-DFT. For KS-DFT, the phonon calculations were conducted in QE using density functional perturbation theory [47,48]. For OF-DFT, the *dftpy* package was used as a calculator within the atomic simulation environment (ASE) Python package to compute phonons via finite differences, employing $5 \times 5 \times 5$ supercells. Due to the low computational cost of OF-DFT simulations with *dftpy* [49], the OF-DFT phonon calculations did not add a significant overhead.

Isolated atom calculations were performed by placing one atom in a cubic cell with a side length of 15 Å.

V. RESULTS AND DISCUSSION

The new set of pseudopotentials generated by the procedure in Sec. III are termed PGBRV1.0, PPSL1.0, PGBRV0.2, and PPSL0.2 where the prepending “P” represents the pseudization for OF-DFT, PSL and GBRV represent the ref-

erence KS NLPP used, and the digits represent the kind of KEDF used to build Δv_{LPP} . We employed the Thomas-Fermi (TF) plus von Weizsäcker (vW) functional with 100% vW (1.0) or 20% vW (0.2).

In the following sections, we analyze the performance of the new set of pseudopotentials for transition metal elements. We first analyze the electron density of periodic systems and assess the ability of the new LPPs to reproduce the KS-DFT reference density of the most stable phase of transition metals. We also show that the new LPPs are applicable to several crystal phases not only the ones used for the PP development. We then move to analyze the electron density of isolated atoms to infer on the applicability of the PPs in combination with approximate KEDFs. For selected elements, we also test phonon spectra and an analysis of the electron density for cluster systems.

A. Generation of the new LPPs

The LPPs are generated according to Eqs. (16)–(19) for all transition metals. Inspired by bulk local pseudopotentials (BLPs) [25], we pseudized atoms in the condensed phase, choosing the most stable phases for each of the elements employing the experimental lattice constants.

The success of the pseudization is given by how well the density difference between the target KS-DFT, $n_{\text{KS}}[\hat{v}_{\text{NLPP}}]$, and the OF-DFT, $n_{\text{OF}}[v_{\text{LPP}}]$, pseudodensities in Eq. (18) can be minimized as a function of the parameters [a to e in Eq. (17)] defining the LPP correction $\Delta v_{\text{LPP}}(r)$. We choose the percentage density deviation Δ_v as a measure of success,

$$\Delta_v = 100 \times \frac{\Delta n}{N_v}, \quad (20)$$

where N_v is the number of valence electrons. We note that $\Delta n \leq N_v$, thus $\Delta_v \leq 100$. In Supplemental Material Fig. S1 [37] we show the spatial distribution of the electron density difference between the OF and KS, using a representative cut plane and line cut for the most stable phases for Ag, Cu, and Pd. The figure validates the convergence criteria in Eq. (20).

Figure 2 provides a visual showing three sets: the colors green, red, and blue; green highlights elements where the new pseudopotentials have Δ_v lower than 5%, red elements have Δ_v between 5% and 8%, and blue elements have Δ_v greater than 8%. The figure clearly shows that PGBRV LPPs generally yield pseudodensities closer to the KS reference compared to the other LPPs. We note that the pseudization dramatically improves upon the densities determined using directly the local part of the NLPP (i.e., $v_{\text{loc}}^{\text{nl}}$) which show an erratic behavior with often very inadequate Δ_v values. This is an important result as using the local part of the NLPP is a common practice in OF-DFT whenever accurate LPPs are not available [50].

We observed that for those elements where the local part of the NLPP already provides pseudodensities $\sim 15\%$ deviated from the KS-DFT reference, minimization of Eq. (18) is challenging. We observed this to be the case for Nb, Mo, and Hg for PGBRV0.2, and Zr, Nb, Y, and Mo for PGBRV1.0.

When comparing PGBRV0.2 (PGBRV1.0) with PPSL0.2 (PPSL1.0), we observe that elements pseudized well by one are often less effectively pseudized by the other. Our analysis

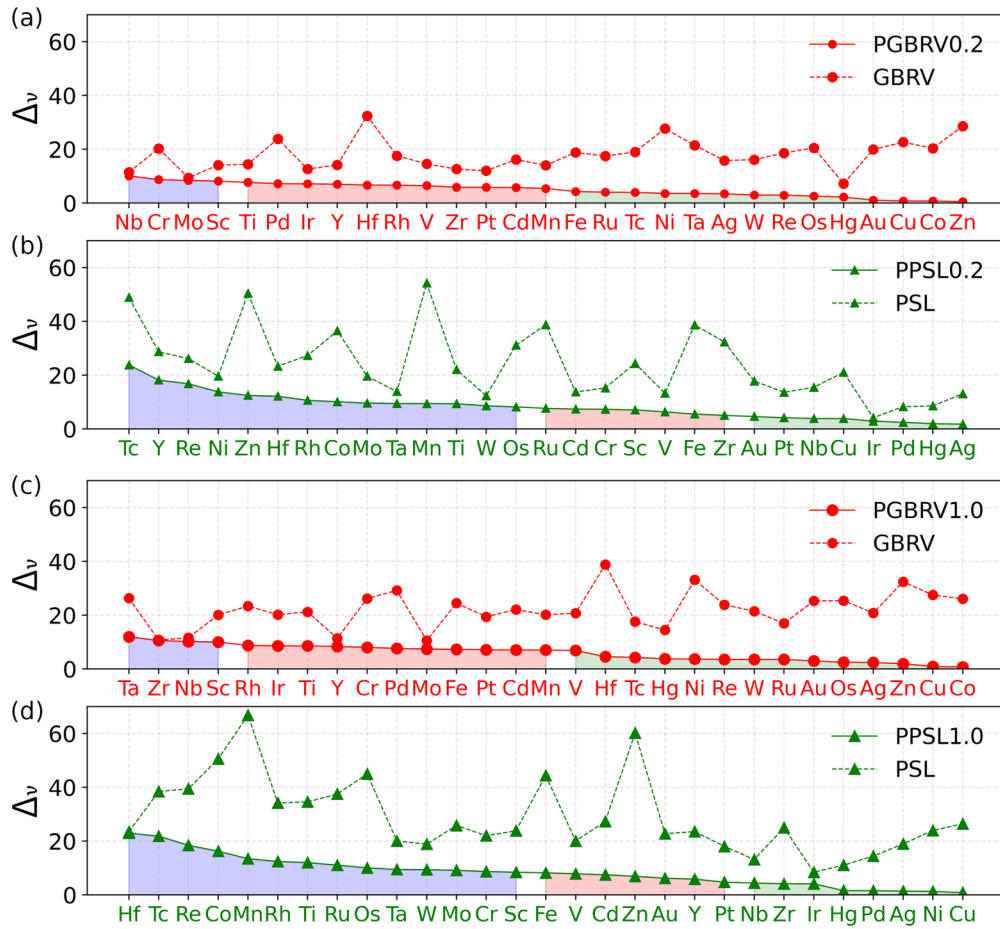


FIG. 2. Percent density deviation, Δ_v , for the atoms in the most stable phase of all transition metals. Green regions are elements with $\Delta_v \leq 5\%$; red $5 < \Delta_v \leq 8\%$, and blue $\Delta_v > 8\%$. The four panels are each devoted to one set of pseudopotentials (PGBRV0.2, PGBRV1.0, PPSL0.2, PPSL1.0, see text for more details).

suggests this behavior stems from differences in the cutoff radii of the two NLPPs, which directly affect the domain of the polynomial defined in Eq. (17). Specifically, as illustrated in Supplemental Material Table S1 and the EOS results in Tables S2–S5 [37], when the GBRV $R_{\text{cut}}^{\text{nl}}$ is larger than that of PSL (e.g., for Nb), PPSL0.2 [recalling our choice $r_{\text{cut}} = \max(R_{\text{cut}}^{\text{nl}})$ in Eq. (17)] results in smaller density deviations compared to PGBRV0.2. Conversely, for elements such as Ni, Re, Zn, Ta, and W, the smaller GBRV $R_{\text{cut}}^{\text{nl}}$ yields more accurate pseudization with PGBRV0.2 than with PPSL0.2. Similarly, PGBRV1.0 pseudopotentials outperform PPSL1.0 for Re and W, while PPSL1.0 provides better results for Zr. These observations underscore the significance of r_{cut} in defining LPPs, suggesting that the optimal choice of cutoff radius should be element specific and optimized.

In sum, PGBRV0.2 and PGBRV1.0 LPPs provide a better pseudization than PPSL0.2 and PPSL1.0. Elements successfully pseudized by PGBRV0.2 and PGBRV1.0 are Co, Cu, Zn, Os, Ni, Ag, Au, Re, W, Hg, Ru, and Tc, while for Pd, Cd, V, Mn, and Pt the pseudization is moderately successful. Nb and Sc are consistently difficult elements for both TFvW and TF0.2vW functionals and NLPPs.

To understand (and predict) the quality of the pseudization, we consider the density displacement presented just now but we order the data by element group and split the data for

the three transition metal periods. We also change slightly the plotted indicator to the density deviation relative to the number of d electrons (rather than valence electrons, see Fig. 3), which we define as follows:

$$\Delta_d = 100 \times \frac{\Delta n}{N_d}, \quad (21)$$

where Δn is defined in Eq. (18), and N_d is the number of valence d electrons. Δ_d normalizes Δn differently from Eq. (20). Specifically, N_v includes semicore electrons while N_d only includes valence d electrons. Some elements may be pseudized differently by the GBRV and PSL NLPPs. For example Ag with GBRV has $4s$ and $4p$ semicore electrons (total of 8) while PSL includes no semicore electrons in the pseudodensity of this element.

A general trend arises from Fig. 3: Δ_d varies monotonically along the transition metal rows. That is, for late transition metals (e.g., Cu, Zn, Ag, Cd, Au, and Hg) both types of LPPs converge to similar, low Δ_d values indicating good pseudization. In contrast, early transition metals (e.g., Sc, Ti, and Y) exhibit larger displacements for both LPPs. We interpret this behavior noticing that KEDFs are known to perform better as the number of electrons increases, consistent with the exactness of the Thomas-Fermi functional in the limit of an infinite electron system [10].

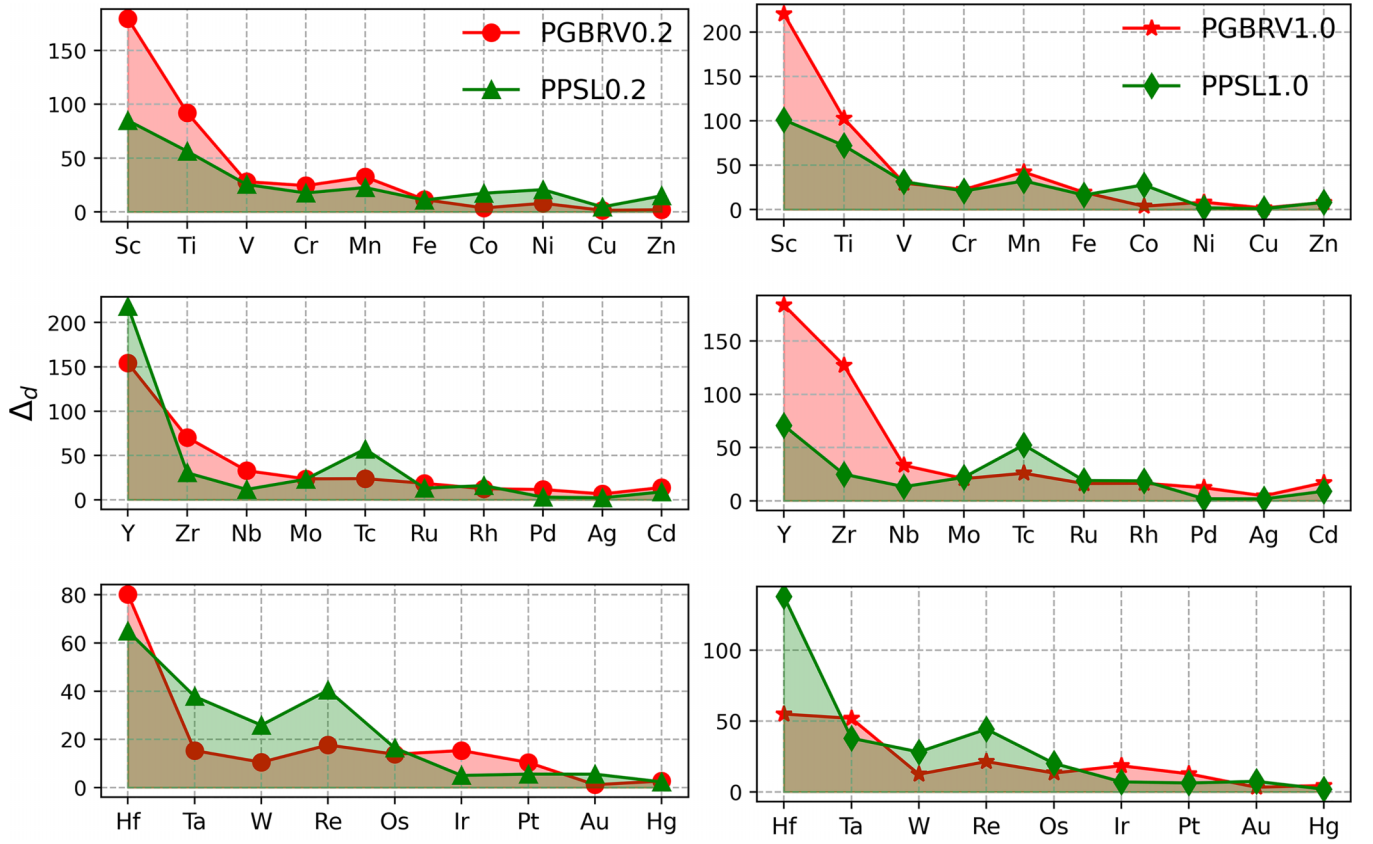


FIG. 3. Percent density deviation per d electron, Δ_d . The panels compare pseudopotentials constructed using the TF0.2vW functional (left) and the TFvW functional (right) for transition metals. Red circles represent GBRV pseudopotentials, green triangles PSL.

We also observe considerable variability in results for group 3–5 transition metals, independent of the period. For example, the PGBRV pseudopotentials yield higher displacements for Sc and Ti compared to PPSL, whereas for Y, PPSL1.0 performs best, followed by PPSL0.2 and then the PGBRV LPPs. While the exact reasons for this behavior remain unclear (we note that GBRV and PSL pseudize each of Y, Sc, and Ti with the same number of valence electrons), as noted before it likely relates to the different r_{cut} values employed in these LPPs. For Y, GBRV uses $r_{\text{cut}} = 2.2 a_0$, while PSL uses $r_{\text{cut}} = 1.8 a_0$. This highlights the need for a followup study focused on optimizing r_{cut} for these new pseudopotentials. Although we have adopted r_{cut} values from the original NLPPs, more suitable choices may exist.

B. Equation of state

From the previous analysis, we have chosen to further investigate PGBRV0.2 for a selected set of atoms such as Cu, Ag, and Pd for which PGBRV0.2 is expected to improve upon the local part of the GBRV PP (i.e., the $v_{\text{loc}}^{\text{nl}}$ mentioned before). We also present Mo for which we predict the PGBRV0.2 will not provide improved electronic structure compared to the local part of the GBRV NLPP. The analysis involves computation of EOS parameters (equilibrium volume, V_0 ; equilibrium energy, E_0 ; and bulk modulus, B_0) for face-centered-cubic (FCC), body-centered-cubic (BCC), and hexagonal-close-packed (HCP) phases. We compare the

PGBRV0.2 results with the local part of the GBRV NLPP and HQLPP. All calculations are carried out with TF0.2vW and the nonlocal revHC KEDFs. The results are collected in Tables I for Ag, II for Cu, III for Pd, and IV for Mo, where we show the relative error of V_0 and B_0 and the energy difference compared to the most stable phase of the presented elements. The EOS parameters from KS (LDA) are compared to experimental values in Table S9 showing that the KS-DFT benchmark results are close to experimental ground truth for lattice constants and within 30% for bulk moduli.

Generally, we notice that PGBRV offers an improved description compared to the local part of GBRV regardless of the functional employed. In comparison to HQLPP, the PGBRV LPPs generally perform better. For example, they provide better V_0 and better B_0 . This is true generally except perhaps for Mo. We notice that for Mo the Δ_v metric for pseudization did not show any improvement compared to the local part of GBRV [see Fig. 2(a)].

Analyzing the specifics, we see that while the EOS parameter values for either TF0.2vW or revHC are not quantitative, they recover important trends for the equilibrium volume (even though it is generally underestimated by about 10%–30%) as well as the bulk modulus. For the latter, we note that approximate KEDFs generally struggle to produce qualitative bulk moduli. However, despite the large deviation, we can discern a generally recovered trend. For example, HCP B_0 is systematically larger than the FCC or BCC B_0 . The energy ordering of the phases is also generally

TABLE I. Percentage errors in the equation of state parameters for V_0 and B_0 with respect to KS-DFT results, and energy difference against the stable phase E_0 of Ag. A dash indicates that the phase is found to be unbound.

Ag	Functional	PP	V_0 (%)	E_0 (eV/atom)	B_0 (%)
FCC	KS	GBRV	0.00	-4023.32	0.00
		GBRV	+43.0	-3719.80	-44.6
	TF0.2vW	PGBRV0.2	-20.1	-4546.42	-22.3
		HQLPP	+34.7	-4054.91	-50.4
		GBRV	+58.1	-3731.58	-56.8
	revHC	PGBRV0.2	-13.3	-4380.85	-12.2
		HQLPP	+22.6	-3998.92	-50.4
HCP	KS	GBRV	0.00	0.01	0.00
		GBRV	+35.8	0.14	-59.5
	TF0.2vW	PGBRV0.2	-17.2	0.04	-30.1
		HQLPP	+33.9	0.01	-66.5
		GBRV	-	-	-
	revHC	PGBRV0.2	-14.1	0.01	-27.7
		HQLPP	+19.0	0.01	-35.8
BCC	KS	GBRV	0.00	0.04	0.00
		GBRV	+42.8	0.05	-45.7
	TF0.2vW	PGBRV0.2	-20.7	0.04	-25.4
		HQLPP	+34.7	0.00	-49.3
		GBRV	-	-	-
	revHC	PGBRV0.2	-14.4	0.0	-10.1
		HQLPP	+34.5	-0.02	-55.8

recovered with FCC consistently predicted to be the most stable phase.

Even though the results presented in Tables I–IV, as well as in Supplemental Material Tables S2–S5 [37], are not quantitatively accurate, they are nonetheless reasonable. In Table

S6 we show a statistical summary of Tables S2–S5, where the mean absolute relative error (MARE) values obtained for V_0 and B_0 using the original GBRV and PSL pseudopotentials indicate high deviations between the OF-DFT and KS-DFT results. In contrast, the new local pseudopotentials

TABLE II. Percentage errors in the equation of state parameters for V_0 and B_0 with respect to KS-DFT results, and energy difference against the stable phase E_0 of Cu. A dash indicates that the phase is found to be unbound.

Cu	Functional	PP	V_0 (%)	E_0 (eV/atom)	B_0 (%)
FCC	KS	GBRV	0.00	-5490.58	0.00
		GBRV	+81.0	-4613.32	-63.5
	TF0.2vW	PGBRV0.2	+11.5	-6947.80	-56.1
		HQLPP	+78.5	-5599.68	-73.5
		GBRV	+115.0	-4625.80	-66.7
	revHC	PGBRV0.2	+27.2	6979.97	-75.7
		HQLPP	-	-	-
HCP	KS	GBRV	0.00	0.02	0.00
		GBRV	-	-	-
	TF0.2vW	PGBRV0.2	+8.8	0.01	-55.8
		HQLPP	+103.2	0.02	-94.4
		GBRV	-	-	-
	revHC	PGBRV0.2	-8.3	-0.038	-56.7
		HQLPP	+84.4	-5483.58	-52.8
BCC	KS	GBRV	0.00	0.05	0.00
		GBRV	+81.3	0.02	-64.5
	TF0.2vW	PGBRV0.2	+11.1	0.01	-55.9
		HQLPP	+116.3	0.00	-98.4
		GBRV	+144.0	0.07	-86.6
	revHC	PGBRV0.2	+24.8	0.00	-73.7
		HQLPP	-	-	-

TABLE III. Percentage errors in the equation of state parameters for V_0 and B_0 with respect to KS-DFT results, and energy difference against the stable phase E_0 of Pd. A dash indicates that the phase is found to be unbound.

Pd	Functional	PP	V_0 (%)	E_0 (eV/atom)	B_0 (%)
FCC	KS	GBRV	0.00	-2704.79	0.00
		GBRV	+68.8	-2336.16	-69.7
	TF0.2vW	PGBRV0.2	+14.5	-2929.42	-61.8
		HQLPP	+55.2	-3544.12	-76.3
		GBRV	+51.6	-2343.41	-51.3
	revHC	PGBRV0.2	-25.2	-2927.97	+7.9
		HQLPP	+31.5	-3488.77	-71.5
HCP	KS	GBRV	0.00	0.03	0.00
		GBRV	+58.8	0.20	-84.6
	TF0.2vW	PGBRV0.2	+11.0	0.00	-59.2
		HQLPP	+43.6	.01	-68.9
		GBRV	+45.2	0.061	-66.3
	revHC	PGBRV0.2	-26.4	-0.006	-13.9
		HQLPP	-	-	-
BCC	KS	GBRV	0.00	0.05	0.00
		GBRV	+67.7	0.02	-67.3
	TF0.2vW	PGBRV0.2	+14.1	0.02	-60.5
		HQLPP	+52.4	0.00	-73.2
		GBRV	+47.8	-0.08	-43.6
	revHC	PGBRV0.2	-26.6	0.00	+14.1
		HQLPP	+22.0	0.00	-60.9

(PGBRV0.2 and PPSL0.2) significantly reduce the MARE for V_0 to 28%–37%, corresponding to a significant improvement in accuracy relative to employing the local part of the KS-DFT pseudopotentials. Although the B_0 values remain less accurate (MARE $\approx$ 55%–60%), all materials are predicted to be

bound. These results are nontrivial for OF-DFT calculations [51,52], and as previously noted, the OF-DFT calculations employing the newly developed LPPs successfully capture several trends observed in the KS-DFT EOS parameters. To rationalize this outcome, we emphasize that the PGBRV0.2

TABLE IV. Percentage errors in the equation of state parameters for V_0 and B_0 with respect to KS-DFT results, and energy difference against the stable phase E_0 of Mo. A dash indicates that the phase is found to be unbound.

Mo	Functional	PP	V_0 (%)	E_0 (eV/atom)	B_0 (%)
BCC	KS	GBRV	0.00	-1897.97	0.00
		GBRV	+22.3	-2140.32	-72.4
	TF0.2vW	PGBRV0.2	+38.1	-1990.97	-74.8
		HQLPP	+40.9	-1906.13	-77.3
		GBRV	-3.2	-2145.08	-47.9
	revHC	PGBRV0.2	+19.7	-1994.48	-62.2
		HQLPP	-11.3	-1863.92	-9.8
FCC	KS	GBRV	0.00	0.43	0.00
		GBRV	+20.4	-0.02	-69.6
	TF0.2vW	PGBRV0.2	+34.7	0.04	-74.5
		HQLPP	+38.4	0.00	-74.1
		GBRV	-2.3	0.11	-45.6
	revHC	PGBRV0.2	+19.9	0.00	-59.3
		HQLPP	-11.8	0.01	-5.7
HCP	KS	GBRV	0.00	0.44	0.00
		GBRV	+17.2	0.01	-69.6
	TF0.2vW	PGBRV0.2	+30.0	0.04	-73.7
		HQLPP	+36.9	0.00	-78.2
		GBRV	-1.5	0.02	-56.4
	revHC	PGBRV0.2	+18.5	0.0	-65.7
		HQLPP	-12.1	0.0	-23.4

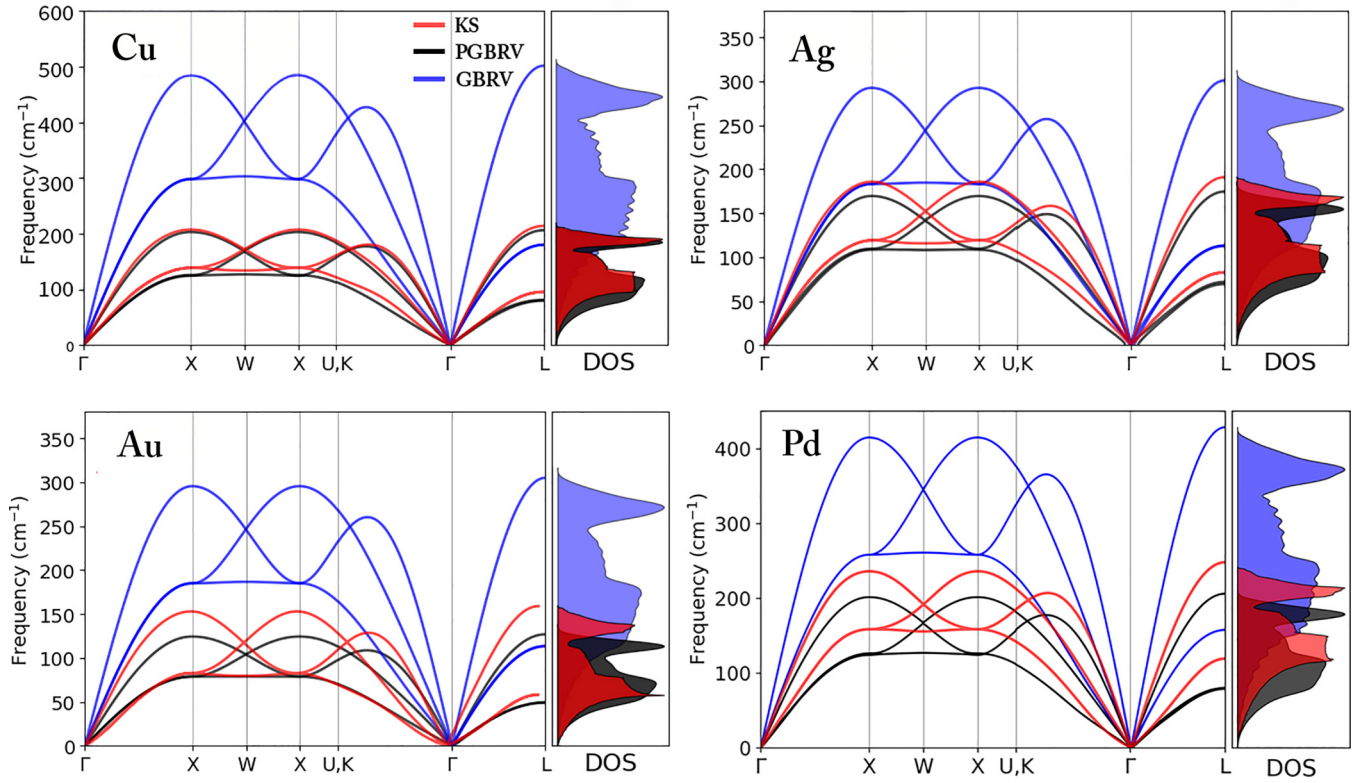


FIG. 4. Phonon spectra of Cu, Ag, Au, and Pd calculated with OF-DFT using TF0.2vW with PGBRV0.2 LPPs (black lines) and the local part of the GBRV NLPPs (blue lines). Reference KS-DFT are also included (red line).

LPP is designed not only to partially recover effects associated with the nonlocal component of the NLPP, but also, crucially, to locally invert the KS density in regions near the ionic cores. Consequently, using the PGBRV0.2 pseudopotentials yields electron densities of substantially higher quality than would be obtained using a generic, KEDF-agnostic LPP. A clear demonstration of this point is provided by the performance comparison with HQLPP pseudopotentials, which yield lower accuracy than PGBRV0.2. A more detailed quantitative comparison between the performance of PGBRV0.2 and HQLPP pseudopotentials is presented in Table S7, where statistical metrics are reported for the FCC, BCC, and HCP phases of Ag, Cu, Pd, and Mo using the TF0.2vW functional. When switching to the revHC functional (Table S8), the PGBRV0.2 pseudopotentials retain a similar level of accuracy (MARE $\approx 20\%$ – 40%), whereas the HQLPPs show noticeably larger deviations. These results confirm that the new pseudopotentials achieve improved consistency with respect to KS-DFT and additionally that their improved performance over using the local part of the KS-DFT PPs is retained across various KEDFs. This discrepancy does not reflect a flaw in the HQLPP pseudopotentials, as they were explicitly constructed to best represent electron densities calculated with the exact KEDF.

C. Phonons

Figure 4 shows the phonon spectra for the noble metals in their FCC phase (Cu, Ag, and Au) as well as Pd also in the FCC phase calculated with TF0.2vW and two LPPs: the

local part of GBRV and PGBRV0.2. The reference KS-DFT with GBRV NLPP is also included. Despite noticing a general underestimation, phonon dispersions produced by PGBRV0.2 are in good agreement with KS-DFT results and are seen to improve dramatically on the results from using the LPP from the local part of the GBRV NLPP.

From the previous sections we found that although Pd is an element that does not reach high convergence in its pseudization, we found PGBRV0.2 still delivered physical EOS parameters. For this reason, we show its phonon spectrum in the last panel of Fig. 4. We note that Pd phonons computed with the PGBRV0.2 LPP are once again much improved compared to using the local part of the GBRV NLPP. In the Supplemental Material [37] we show the phonon dispersion of Mo and Zr where the local part of GBRV produces several imaginary frequencies, PGBRV0.2 does not. Although for these elements the comparison with KS-DFT is not as clean as for the noble metals or Pd, the improved phonon dispersion of PGBRV0.2 vs local part of GBRV is still quite clear. These observations are consistent with our analysis of the pseudization in Sec. V A, Zr is the worst performer.

D. Compatibility of KEDF and LPP combinations

To evaluate the compatibility and performance of the proposed LPPs with several KEDF approximants, we examine the electron density of isolated atoms between two classes of KEDFs: semilocal (TFvW, TF0.2vW, LKT [53], revAPBEK [9]) and nonlocal (LMGP and revHC) KEDFs in comparison with densities computed with either TF0.2vW or TFvW

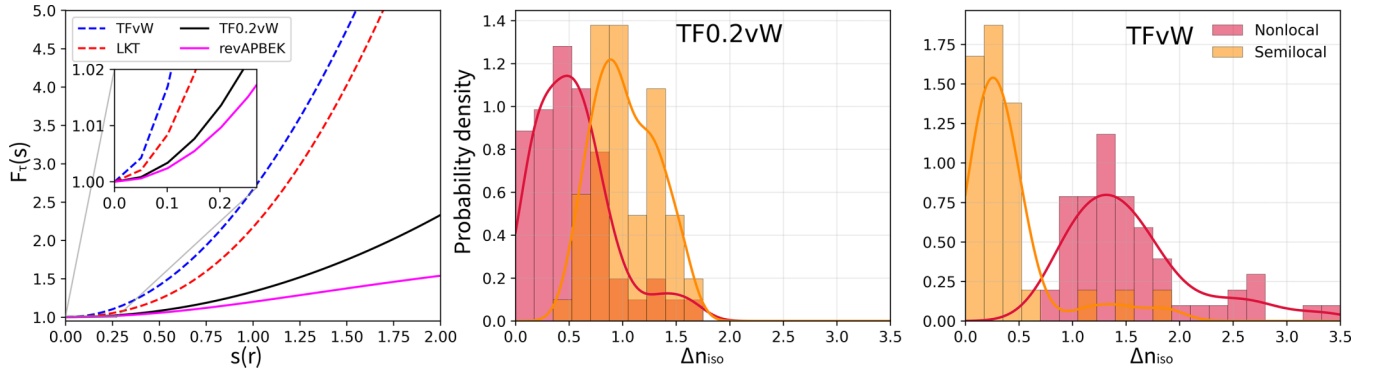


FIG. 5. Leftmost panel: enhancement factor $F_r(s)$ of several GGA functionals. Right two panels: electron density deviation Δn_{iso} against TF0.2vW and TFvW functionals for several GGA KEDF (LKT, revAPBEK, TFvW) and nonlocal functionals (revHC and LMGP).

employing the new LPPs (see Fig. 5). To better understand the results for semilocal KEDFs, we consider the enhancement factor $F(s)$ as a function of the reduced density gradient $s(\mathbf{r})$. Further details on the explicit forms of $F(s)$ for the various semilocal KEDFs (TFvW, TF0.2vW, LKT, and revAPBEK) are provided in the Supplemental Material, Sec. II [37]. The leftmost panel of Fig. 5 shows $F(s)$ where we notice that for small values of $s(\mathbf{r})$, where the electron density is usually large LKT, revAPBEK, and TFvW have a noticeably higher $F(s)$ compared to TF0.2vW. The enhancement factor for TF0.2vW, due the small vW prefactor of 0.2, displays a slower upward behavior compared to the other functionals. Similarly for large values of $s(\mathbf{r})$, which correspond to diffuse or tail regions of the density, the enhancement factor increases significantly for TFvW and LKT while it remains low for TF0.2vW and revAPBEK. The latter is due to its asymptotic behavior [9]. This analysis shows that LKT, revAPBEK and TFvW are likely more compatible with densities form TFvW in conjunction with PGBRV1.0 and PPSL1.0 than with densities from TF0.2vW in conjunction with PGBRV0.2 and PPSL0.2.

The two histograms of Δn_{iso} [defined in Eq. (18) but applied to isolated atoms] in Fig. 5 clearly show that non-local functionals yield densities closer to TF0.2vW (with PGBRV0.2) with an average deviation of $0.5e$. In contrast,

these functionals against TFvW (with PGBRV1.0) show an average deviation of $1.4e$. Semilocal functionals show an opposite behavior deviating by $1.0e$ from TF0.2vW and a meager $0.25e$ for TFvW.

Overall, these results emphasize the importance of internal consistency between the chosen LPP and the KEDF, particularly when applying OF-DFT to systems characterized by large density gradients or strong localization effects. This compatibility issue between KEDF and LPP was seen as a negative aspect by some [25]. However, in the context of this work, we see that already the two functionals considered for LPP construction, TFvW and TF0.2vW, form an excellent platform or “basis set” of KEDFs to be able to accommodate an array of functionals. Should one require a functional-specific LPP, the option to follow the LPP construction workflow presented here is always available and achievable using the software provided in the Supplemental Material [37].

E. Clusters

In this section, we evaluate the performance of the new PGBRV0.2 LPPs in describing the electronic density of finite systems. Specifically, we study Pd_{13} , Ag_{20} , and Pt_{10} clusters with results summarized in Tables V–VII (for PPSL see Supplemental Material Tables S6–S8 [37]). The electronic density difference Δn is computed as in Eq. (18). From the results, we observe that the electron density calculated with

TABLE V. Relative density difference (against KS) for Pd_{13} . The OF-DFT electron density is computed with the local part of the GBRV NLPP as well as with PGBRV0.2. Improvement (%) indicates the percent improvement of the PGBRV LPP compared to the local part of GBRV.

System	KEDF	PP	$\Delta n/N_v$	Improvement (%)
Pd_{13}	TF0.2vW	GBRV	0.291	75
		PGBRV0.2	0.074	
	TFvW	GBRV	0.291	56
		PGBRV0.2	0.128	
	LKT	GBRV	0.268	58
		PGBRV0.2	0.113	
	revHC	GBRV	0.240	59
		PGBRV0.2	0.074	
	LMGP	GBRV	0.221	67
		PGBRV0.2	0.073	

TABLE VI. Relative density difference (against KS) for Ag_{20} . Other details as in Table V.

System	KEDF	PP	$\Delta n/N_v$	Improvement (%)
Pd_{13}	TF0.2vW	GBRV	0.182	77
		PGBRV0.2	0.042	
	TFvW	GBRV	0.182	67
		PGBRV0.2	0.059	
Ag_{20}	LKT	GBRV	0.164	77
		PGBRV0.2	0.038	
	revHC	GBRV	0.128	63
		PGBRV0.2	0.047	
	LMGP	GBRV	0.107	22
		PGBRV0.2	0.084	

TABLE VII. Relative density difference (against KS) for Pt₁₀. Other details as in Table V.

System	KEDF	PP	$\Delta n/N$	Improvement (%)
Pt ₁₀	TF0.2vW	GBRV	0.290	79
		PGBRV0.2	0.061	
	TFvW	GBRV	0.290	56
		PGBRV0.2	0.128	
	LKT	GBRV	0.277	62
		PGBRV0.2	0.105	
	revHC	GBRV	0.247	71
		PGBRV0.2	0.071	
	LMGP	GBRV	0.241	72
		PGBRV0.2	0.067	

the PGBRV0.2 LPPs is in generally good agreement with KS electron density, showing an error of 10% or less.

The best performers are LMGP for Pd₁₃ (7.3% deviation), LKT for Ag₂₀ (3.8% deviation), and TF0.2vW for Pt₁₀ (6.1% deviation). When averaging over all three clusters, the TF0.2vW functional yields the smallest overall electron density difference ($\Delta n/N_v = 0.059$), thus representing the best overall performer in this set, followed closely by revHC (0.064), LMGP (0.075), LKT (0.085), and lastly TFvW (0.105). The results also show that the electron density is always improved (up to 79% improvement!) compared to using the local part of the GBRV NLPP.

VI. CONCLUSIONS

In this study, we developed new sets of pseudopotentials for OF-DFT calculations targeting transition metal elements. Transition metals have historically been excluded from OF-DFT applications due to the challenges of constructing suitable local pseudopotentials. The core idea of this work is to adapt existing KS-DFT pseudopotentials (specifically, PSL and GBRV) by imposing that OF-DFT calculations with approximate noninteracting kinetic energy functionals reproduce, as closely as possible, the electron density from KS-DFT calculations of the element's most stable condensed phase. This approach effectively performs a KS-to-OF inversion near the ion cores. Our work opens the possibility to investigate large-scale systems and nanostructures composed of transition metals at low cost (linear scaling with low prefactor).

Our convergence analysis shows that GBRV-based pseudopotentials generally outperform PSL pseudopotentials for

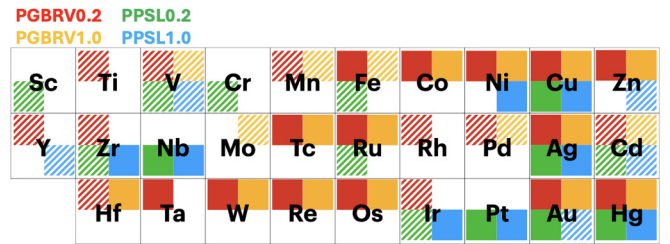


FIG. 6. Recommendations for using the proposed pseudopotentials. The color map illustrates the outcome of the pseudization procedure. Solid colors represent elements for which the pseudopotential yields accurate results; hatched areas indicate lower-quality pseudization; and uncolored elements correspond to pseudopotentials we consider inaccurate. These recommendations are based on three sets of data: the electron density in the bulk system, the electron density of the isolated atom, and equation of state parameters.

most transition metals. Figure 6 summarizes our recommendations for each element: solid colors indicate successful pseudization, while hatched areas denote intermediate accuracy (see figure caption for details).

We conclude with a note of caution: there are still major challenges for applying OF-DFT methods to transition metals. While pseudopotentials are increasingly available (see, e.g., Refs. [19,20,29] and this work), the overall accuracy of existing physics-based OF-DFT methods is still too low for the widespread use of OF-DFT with these important elements. However, the presented pseudopotentials take us a step closer to making OF-DFT a practical tool for transition-metal systems, even if further advances in KEDFs are required to reach chemical accuracy.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The data that support the findings of this article are openly available [62], embargo periods may apply.

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